The first pick was right.
The task was not done.

Grounded Intent | ZHAW: See It, Say It, Pick It

Our approach in one sentence

Apertus interprets the request; image-only vision supplies observations; explicit rules compile and check an ordered pick/place proposal.

The problem we found

A full-credit first pick can hide a missing placement. We audit the complete task, not just the pixel that starts it.

The whole local pipeline on one GPU

One RTX 4070 Ti · 12 GB VRAM | Qwen3.5-9B + Apertus 1.5 8B Q8_0
Sequential model loading; no robot execution.

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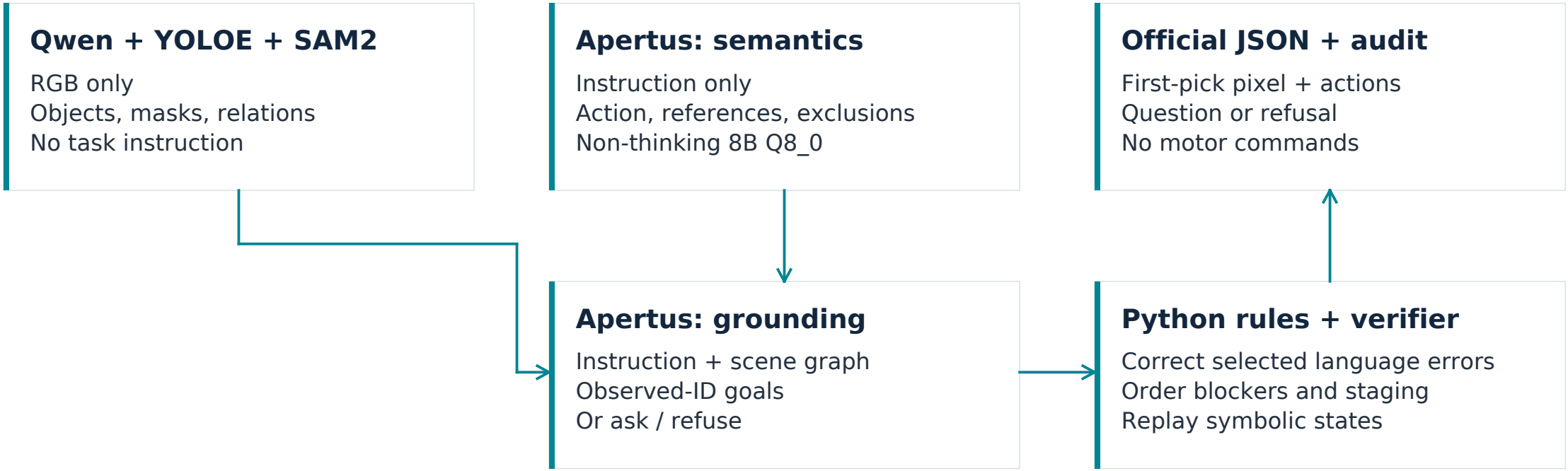
Apertus interprets. Code can still overrule it.

Model roles and the boundary between language and symbolic execution



Downstream rules include category, pickup and spatial interpretation. This is a hybrid system, not model-only reasoning.

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Maximum public score. Not full-task success.

Fresh local benchmark run | 10 scenes, 50 instructions

Grounding 40 / 40 first picks	**35.0 / 35**
Safety response 10 / 10; 0 false alarms	**30.0 / 30**
Automatic total Public development set	**65.0 / 65**

Beyond the development set

The saved held-out file has 40 answers.
Its labels are unpublished;
we report no held-out accuracy.

No robot execution or certified safety is claimed.

What the script measures

Correct first-pick pixel and plan / ask / refuse. It does not score completion of every later action.

What the full plans reveal

One placement request stops after the pick: vision missed the destination tray. A correct first pixel does not make that task complete.

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Where Apertus had difficulty

Observed Apertus failures | local probes and separately labelled remote diagnostics

Smaller questions helped

Local full parser: three invalid replies. A relation-only call recovered in/on from 1/2 to 2/2. Focused calls passed 3/3 simple controls, with quantity supplied by the harness. [E1, E2]

Apertus behind the camera

Remote Apertus vision + YOLOE/SAM2: 37.625/65; saved Qwen: 65/65. First picks: 16/40; non-plans: 9/10; false asks/refusals: 5/40. Local Apertus language reasoning. [E10]

A fix that stayed with its examples

Remote clarification: 9/10 to 10/10 on matched controls. On fresh wording, both prompts scored 7/10. The transfer test found no gain. [E3]

A substitute instead of a refusal

Remote 70B chose blue panels for a missing blue cube. Local 8B refused. We observed the substitution; its internal cause is unknown. [E4]

Remote CSCS probes are development diagnostics, non-compliant as submission measurements under the published local-only rule.

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What we would keep. What we would test next.

Lessons from the experiments | Next steps before physical execution

Keep Qwen on vision

The drop-in vision test favoured Qwen in our pipeline. Keep Apertus on language. A mask-interior point still needs to belong to the object the user requested.

Check the original request

Keep targets, quantities and destinations independently of the proposed plan. Otherwise a verifier can approve a task the user never asked for.

Test beyond the repaired example

Use fresh paraphrases, negative controls and missing destinations. A prompt change that fixes its motivating cases has not yet earned a place in the pipeline.

Rerun the shipped configuration

python run.py --data <scenes> --out results.json
Regenerate answers from the exact shipped tree. Keep symbolic checks separate from physical grasp and motion validation.

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